

Class 10

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Structural Bioinformatics (Pt.1)

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

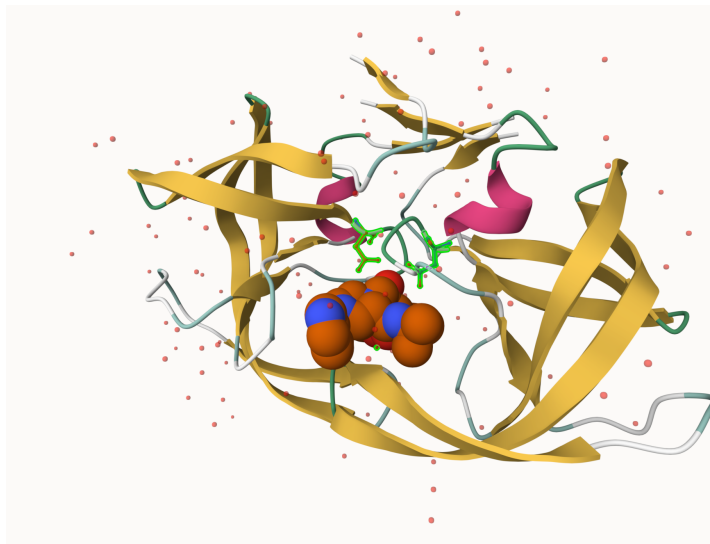
80.5% of the structures in PDB are solved by X-Ray and 13.4% are solved by Electron Microscopy.

Q2: What proportion of structures in the PDB are protein? 97.9% of the structures in PDB contain protein

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,227 HIV-1 protease structures in the current PDB

3. Visualizing the HIV-1 protease structure



Saving an Image

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We see just one atom per water molecule in this structure because the water molecules are represented by a single oxygen atom in the structure. In X-Ray crystal structures only one oxygen atom is normally visible since hydrogen atoms are smaller and harder to detect.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The conserved water molecule in the binding site is HOH308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site?

Larger ligands such as indinavir can access the binding site through conformational changes in this HIV-1 protease. The flap regions of the enzyme transiently open which can create space for the ligand to enter the active site. It can also close to allow proper binding and positioning.

4. Introduction to Bio3D in R

```
#Loading bio3d
library(bio3d)
```

```
#Reading and inspect on-line file with PDB Id 1HSG
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
#Generating summary of the contents of the pdb object
pdb
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this pdb object

Q8: Name one of the two non-protein residues?

One of the two non-protein residues is MK1

Q9: How many protein chains are in this structure?

There are two protein chains in this structure

```
#Finding attributes of any such object
attributes(pdb)
```

```
$names
[1] "atom"    "xyz"     "seqres"  "helix"   "sheet"   "calpha"  "remark"  "call"

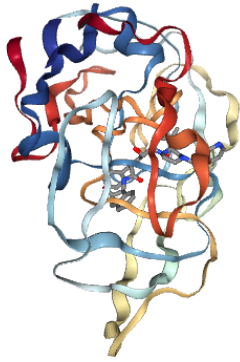
$class
[1] "pdb" "sse"
```

```
#Accessing the atom attribute or component
head(pdb$atom)
```

	type	eleno	ety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

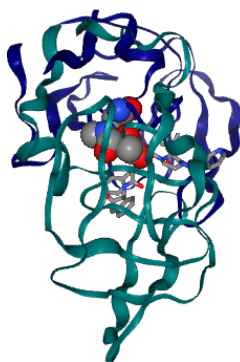
	segid	el	esy	charge
1	<NA>	N		<NA>
2	<NA>	C		<NA>
3	<NA>	C		<NA>
4	<NA>	O		<NA>
5	<NA>	C		<NA>
6	<NA>	C		<NA>

```
#Loading respective packages and generate a quick NGL structure
#overview of a bio3d pdb class object with a number of simple defaults
library(bio3dview)
library(NGLViewerR)
#Viewing the pdb structure
view.pdb(pdb) |>
  setSpin()
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
setRock()
```



Predicting Functional Motions of a Single Structure

```
#Reading a new PDB structure of Adenylate Kinase
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
#Running adk
adk
```

Call: read.pdb(file = "6s36")

Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [CL (3), HOH (238), MG (2), NA (1)]

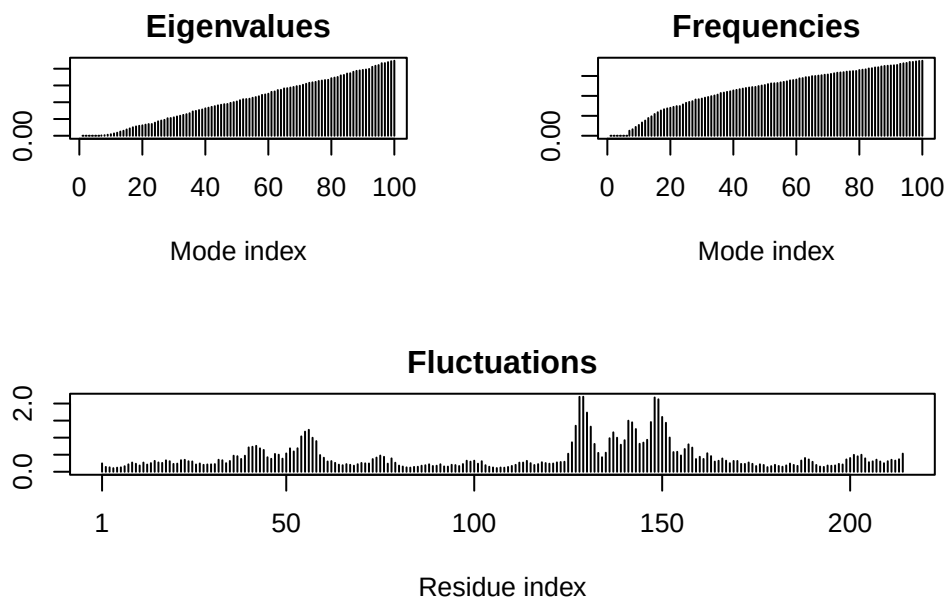
Protein sequence:
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

```
# Perform flexibility prediction
m <- nma(adk)
```

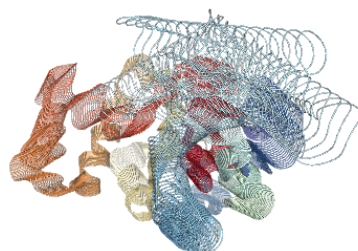
Building Hessian... Done in 0.014 seconds.
Diagonalizing Hessian... Done in 0.292 seconds.

```
#Generating plot for the NMA results
plot(m)
```

```
#Generating molecular "trajectory"  
mktrj(m, file="adk_m7.pdb")
```

```
#Quick display of molecular "trajectory" using view.nma() from bio3dview package  
view.nma(m, pdb=adk)
```



5. Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

The package above that is found only on BioConductor and not CRAN is msa.

Q11. Which of the above packages is not found on BioConductor or CRAN?

The bio3dview package is not found on BioConductor or CRAN.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

Search and retrieve ADK structures

```
#Loading the bio3d package
library(bio3d)
#Retrieving the amino acid sequence for chain A of PDB entry 1ake
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
#Viewing the retrieved sequence
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLVT
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

181 . . . 214

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence

```
#BLAST the ADK sequence against the PDB database to find similar structures
b <- blast.pdb(aa)
```

```
Searching ... please wait (updates every 5 seconds) RID = ZGDFATF016
```

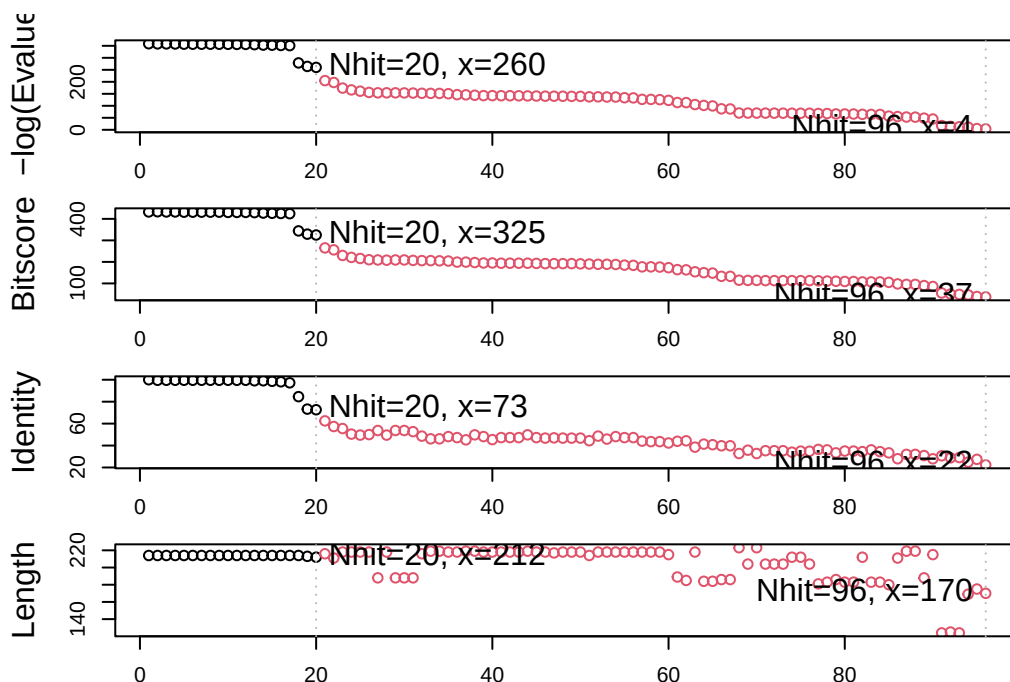
```
.....
```

```
Reporting 96 hits
```

```
#Plot and filter the BLAST results to identify top hits
hits <- plot(b)
```

```
* Possible cutoff values: 260 3
    Yielding Nhits:      20 96
```

```
* Chosen cutoff value of: 260
    Yielding Nhits:      20
```



```
# List out some 'top hits'
head(hits$ pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

```
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6H...
```

```
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6RZE.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download

	0%
=====	8%
=====	15%
=====	23%



```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
```

```
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

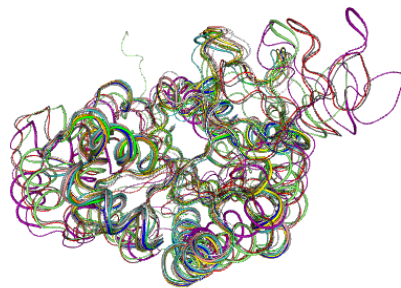
```
..   PDB has ALT records, taking A only, rm.alt=TRUE
....  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

Extracting sequences

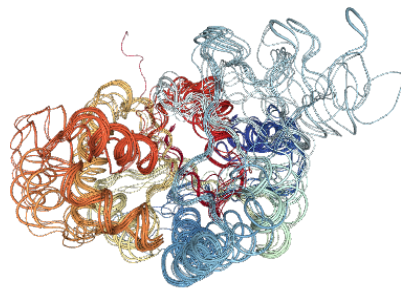
```
pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbs/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbs/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbs/split_chain/4PZL_A.pdb
```

Optional: Viewing our superposed structures

```
#Loading the bio3dview package
library(bio3dview)
#Displaying the retrieved PDB structures in 3D
view.pdbs(pdbs)
```

```
#Loading the bio3dview package
library(bio3dview)
#Displaying the retrieved PDB structures in 3D
view.pdbs(pdbs, colorScheme = "residueIndex")
```



Annotate collected PDB structures

```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
#Viewing available annotation data
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					

3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					

	resolution	scopDomain	pfam
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)
3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,C0
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,PO4
3GMT_A	SO4 (2)
4PZL_A	CA,FMT,GOL

	ligandName
1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)
3HPR_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4V_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A	BIS(ADENOSINE)-5'-

PENTAPHOSPHATE
 3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
 6HAP_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 6HAM_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 4K46_A ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
 3GMT_A SULFATE ION (2)
 4PZL_A CALCIUM ION,FORMIC ACID,GLYCEROL

	source
1AKE_A	Escherichia coli
6S36_A	Escherichia coli
6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
3GMT_A	Burkholderia pseudomallei 1710b
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBIT

6S36_A
 6RZE_A
 3HPR_A
 1E4V_A
 loop
 5EJE_A
 1E4Y_A
 loop
 3X2S_A
 conjugated adenylate kinase
 6HAP_A
 6HAM_A
 4K46_A
 3GMT_A
 4PZL_A

Cryst

	citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560

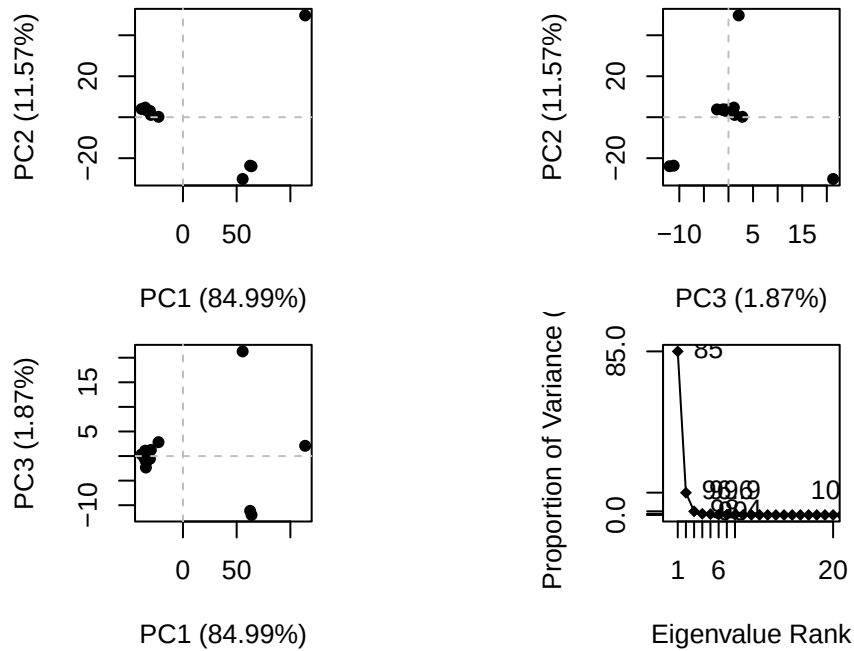
The crys

6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680

rWork spaceGroup

1AKE_A	0.19600	P 21 2 21
6S36_A	0.15940	C 1 2 1
6RZE_A	0.18190	C 1 2 1
3HPR_A	0.20620	P 21 21 2
1E4V_A	0.19600	P 21 2 21
5EJE_A	0.18630	P 21 2 21
1E4Y_A	0.17800	P 1 21 1
3X2S_A	0.20700	P 21 21 21
6HAP_A	0.22370	I 2 2 2
6HAM_A	0.20311	P 43
4K46_A	0.16730	P 21 21 21
3GMT_A	0.23500	P 1 21 1
4PZL_A	0.19130	P 32

```
# Perform PCA
pc.xray <- pca(pdbx)
plot(pc.xray)
```

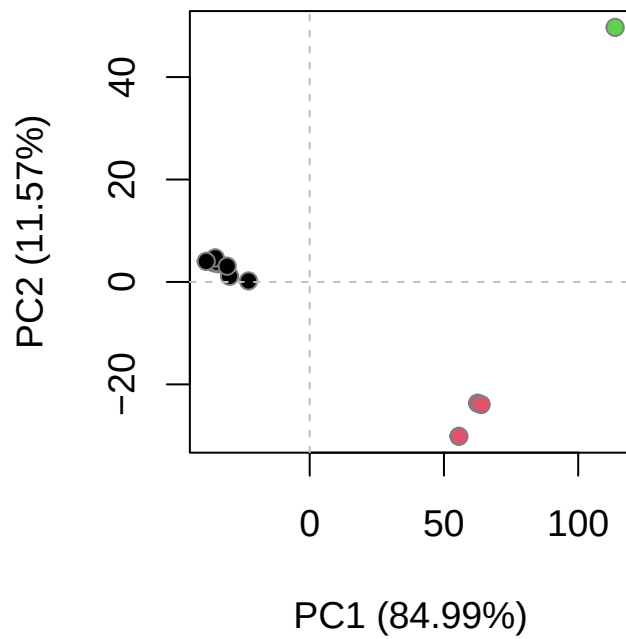


```
# Calculate RMSD
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

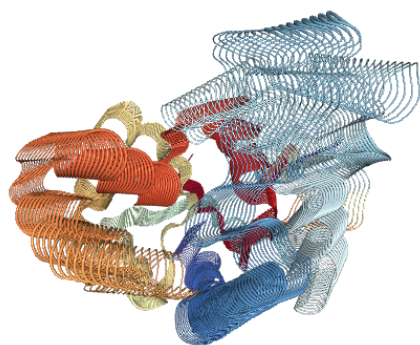
plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```

PCA visualization

```
# Visualize first principal component  
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

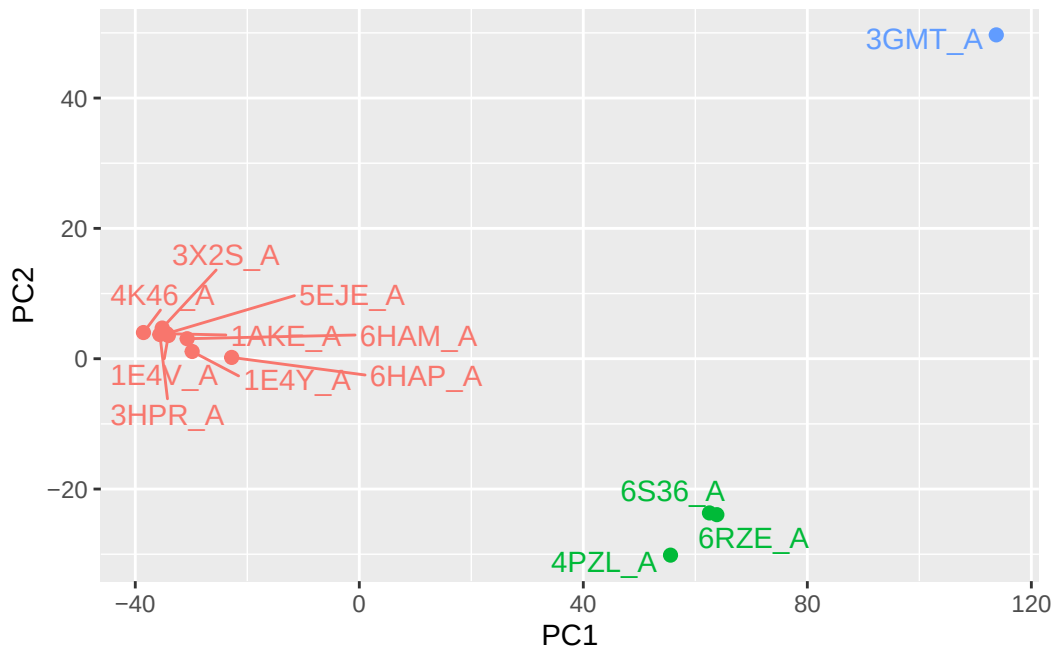
```
#Viewing results with bio3dview view() function  
view.pca(pc.xray)
```



```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                  PC2=pc.xray$z[,2],
                  col=as.factor(grps.rd),
                  ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



6. Normal mode analysis

```
# NMA of all structures
modes <- nma(pdbbs)
```

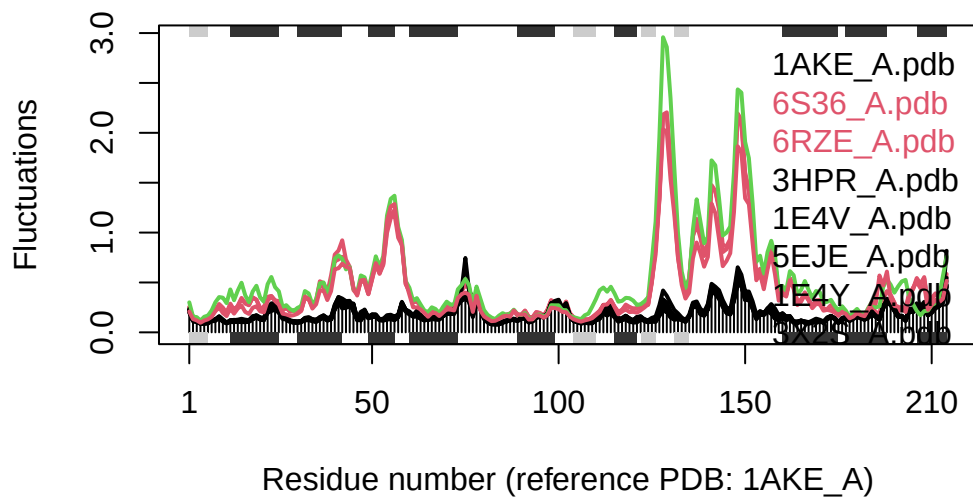
Details of Scheduled Calculation:

```
... 13 input structures
... storing 606 eigenvectors for each structure
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb
```

	0%
=====	8%
=====	15%
=====	23%
=====	31%
=====	38%
=====	46%
=====	54%
=====	62%
=====	69%
=====	77%
=====	85%
=====	92%
=====	100%

```
#Plot fluctuations across residues for all structures
plot(modes, pdbbs, col=grps.rd)
```

Extracting SSE from pdbbs\$sse attribute



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The plot has similar trends, but the red and green lines have higher peaks. The black and colored lines share the same overall shape, but the colored lines have more fluctuations. The colored lines fluctuate more around residues 125-150, which suggests that this region has increased flexibility.